CHAPTER 7

Antiderivatives and differential equations

S52, S53, S55, S58

Objectives

Calculate antiderivatives; solve a differential equation with an initial condition; interpret an Euler approximation.

Course notes

Antiderivatives on an interval

At this stage we accept that every continuous function on an interval has antiderivatives; the link with area will be established in chapter 8. An antiderivative of f on an interval I is a differentiable function F such that $F' = f$ on I. If F is an antiderivative, all the antiderivatives on I are $F + C$, $C \in \mathbb{R}$.

Proof. If $F' = G' = f$, then $(F - G)' = 0$ on the interval; $F - G$ is therefore constant. Conversely, $(F + C)' = f$. The interval hypothesis is essential: on a disconnected domain, the constants may differ from one interval to another.

Table of antiderivatives — to be read as the reverse of derivatives

$f(x)$	$F(x)$
x^n $(n \in \mathbb{Z},\ n \neq -1)$	$\frac{x^{n+1}}{n+1}$
$\frac{1}{x}$	$\ln \lvert x \rvert$ on an interval avoiding 0
$\frac{1}{\sqrt{x}}$	$2\sqrt{x}$ $(x > 0)$
e^x	e^x
$\cos x$	$\sin x$
$\sin x$	$-\cos x$

The additive constant can always be added. The last two rows will be used again after the study of trigonometry.

Recognising a composition

For a differentiable function u, the antiderivatives of $u'u^n$, $u'e^u$, $\frac{u'}{u}$ are respectively $\frac{u^{n+1}}{n+1}$ $(n \neq -1)$, e^u, $\ln \lvert u \rvert$, on the intervals where the expressions make sense. Linearity allows us to handle sums.

Example. An antiderivative of $6x(x^2+1)^2$ is $(x^2+1)^3$; an antiderivative of e^{3x} is $\frac{e^{3x}}{3}$. To impose $F(0) = 2$ in the latter case, take $F(x) = \frac{e^{3x}}{3} + \frac{5}{3}$.

Differential equations

A differential equation relates an unknown function to its derivatives. A solution must be checked on a specified interval.

Standard solutions

$y' = ay$ has solutions $y(x) = Ce^{ax}$. For $a \neq 0$, $y' = ay + b$ has solutions

$$y(x) = Ce^{ax} - \frac{b}{a}.$$

If $a = 0$, the equation becomes $y' = b$ and $y(x) = bx + C$. If g is a known particular solution of $y' = ay + f(x)$, the solutions are $y = g + Ce^{ax}$.

Proof for $y' = ay$. If y is a solution, $z(x) = e^{-ax}y(x)$ satisfies $z' = e^{-ax}(y' - ay) = 0$; $z = C$. Conversely, the derivative of Ce^{ax} is aCe^{ax}. For $y' = ay + b$, subtracting the constant solution $-\frac{b}{a}$ reduces to the homogeneous equation. For a variable right-hand side, do the same with g.

Example. $y' = -2y + 6$ and $y(0) = 1$ give $y = 3 + Ce^{-2x}$, then $C = -2$. The solution $3 - 2e^{-2x}$ increases towards the equilibrium 3 as $x \to +\infty$.

Euler's method

For $y' = ay + b$, with a step $h > 0$, we approximate $y(t_k)$ by Y_k according to $t_{k+1} = t_k + h$, $Y_{k+1} = Y_k + h(aY_k + b)$. This is an approximation using the tangent, not an equality on the exact solution. A smaller step can improve the approximation; check the error numerically on a known solution.

Research activity

Cooling of a drink — 40 min.

A drink initially at 80 °C is placed in a room kept at 20 °C. For $t \geq 0$, let $T(t)$ be its temperature in degrees Celsius, where t is the elapsed time in minutes. We accept the model $T'(t) = -0.1(T(t) - 20)$ and $T(0) = 80$.

1. Solve this initial value problem and give the expression of $T(t)$.
2. Study the direction of variation of T and find its limit at $+\infty$. Interpret the results.
3. Find the first instant at which the temperature is 40 °C. Give the exact value, then an approximate value in minutes.
4. Apply two steps of Euler's method with a step of 1 minute. Compare the approximation obtained after two minutes with the exact value $T(2)$.
5. Give a concrete situation in which the assumption of a constant room temperature would no longer be appropriate.

Progressive exercises

E1 · Practice. Find all the antiderivatives on $\mathbb{R}$ of the function $x \mapsto 3x^2 - 4x + 1$.

E2 · Practice. Find the antiderivative F of the function $x \mapsto \frac{2x}{x^2+4}$ on $\mathbb{R}$ that satisfies $F(0) = 0$.

E3 · Practice. Find the solution on $\mathbb{R}$ of the differential equation $y' = 3y$ that satisfies $y(1) = 2$.

E4 · Practice. Find the solution on $\mathbb{R}$ of the differential equation $y' = -y + 4$ that satisfies $y(0) = 7$.

E5 · Practice. Consider the differential equation $(E) : y' = y - x$ on $\mathbb{R}$. Check that $g(x) = x + 1$ is a solution of it, then find the solution of (E) that satisfies $y(0) = 0$.

Textbook or past-paper work

Barbazo, p. 228, exercises 25 and 30. In S55: exercise 30, to bring out a constant solution. At home: exercise 25 in full, distinguishing a condition on f from a condition on f'.

Homework

HW7 — An evolution with variable input — 60 min.

We look for a function y defined and differentiable on $[0;+\infty[$ such that $y'(t) = -y(t) + 2t + 1$ and $y(0) = 3$.

1. Find two real numbers a and b such that $g(t) = at + b$ is a solution of the differential equation.
2. Deduce the expression of the solution that satisfies $y(0) = 3$.
3. Study the variations of this solution and state its minimum and the instant at which it is attained.
4. Write a Python function with argument N, a strictly positive integer, that applies Euler's method on $[0;2]$ with step $\frac{2}{N}$ and returns the approximation of $y(2)$. Compare with the exact value for $N = 10$, then $N = 100$.
5. Why is checking the differential equality at only a few instants not enough to prove that a function is a solution?

Teaching notes and solutions

Teacher notes

Distinguish the unknown function from the real variable. Have each antiderivative checked by differentiation. For $y' = ay + b$, look first for the constant solution. The case $a = 0$ must remain visible. The trigonometric antiderivatives are given here as a reference table, but their assessed use awaits the trigonometry unit. For Euler, keep the units and compare with a known exact solution.

Solutions to the activity

1. The equation is $T' = -0.1T + 2$. A constant solution is 20; all the solutions are $T(t) = 20 + Ce^{-0.1t}$. Since $T(0) = 80$, $C = 60$. We check $T'(t) = -6e^{-0.1t} = -0.1(T(t) - 20)$ and $T(0) = 80$.

2. $T'(t) = -6e^{-0.1t} < 0$: the temperature decreases strictly. Since $e^{-0.1t} \to 0$ at $+\infty$, $T(t) \to 20\,°\text{C}$, while remaining above $20\,°\text{C}$.

3. $T(t) = 40 \iff e^{-0.1t} = \frac{1}{3}$. Taking the logarithm, $-0.1t = -\ln 3$, that is $t = 10 \ln 3 \simeq 10.9861$ min. Strict decrease ensures that this is the only instant at which the temperature passes through $40\,°\text{C}$.

4. For $h = 1$, $Y_{k+1} = Y_k - 0.1(Y_k - 20)$. Thus $Y_1 = 80 - 6 = 74$ and $Y_2 = 74 - 5.4 = 68.6$. The exact solution gives $T(2) = 20 + 60e^{-0.2} \simeq 69.1238$. Euler underestimates the temperature here by about $0.5238\,°\text{C}$.

5. The model assumes a room temperature of $20\,°\text{C}$ and a constant exchange coefficient of $0.1\,\text{min}^{-1}$. A change of room, a lid or stirring can alter these assumptions; the formula is then no longer automatically appropriate.

E1. Linearity gives $F(x) = x^3 - 2x^2 + x + C$, $C \in \mathbb{R}$. We check $F'(x) = 3x^2 - 4x + 1$. All the antiderivatives on the interval $\mathbb{R}$ differ by a constant.

E2. The denominator is positive and its derivative is $2x$. The antiderivatives are $\ln(x^2 + 4) + C$. The condition at 0 gives $C = -\ln 4$, so $F(x) = \ln(x^2 + 4) - \ln 4$; we check $F(0) = 0$ and $F'(x) = \frac{2x}{x^2+4}$.

E3. The solutions of $y' = 3y$ are Ce^{3x}. The condition $Ce^3 = 2$ forces $C = 2e^{-3}$. Thus $y(x) = 2e^{3(x-1)}$, whose derivative is $3y(x)$ and whose value at 1 is 2.

E4. The constant 4 is a particular solution. The solutions are therefore $4 + Ce^{-x}$. Since $4 + C = 7$, $C = 3$. The solution is $y(x) = 4 + 3e^{-x}$; its derivative $-3e^{-x}$ is indeed $-y + 4$, and $y(0) = 7$.

E5. $g(x) = x + 1$ satisfies $g' = 1 = g - x$. If $z = y - g$, then $z' = z$, so $y = x + 1 + Ce^x$. The condition $y(0) = 0$ forces $C = -1$. We obtain $y = x + 1 - e^x$ and we check $y' = 1 - e^x = y - x$.

Solutions to HW7

1. If $g(t) = at + b$, then $g' = a$ and $-g + 2t + 1 = (2 - a)t + 1 - b$. For the identity to hold for every t, we need $2 - a = 0$ and $a = 1 - b$. So $a = 2$, $b = -1$, and $g(t) = 2t - 1$.

2. Let $z = y - g$. Subtracting the two equations, $z' = -z$, so $z = Ce^{-t}$. Thus $y(t) = 2t - 1 + Ce^{-t}$. Since $y(0) = 3$, $C = 4$. The function $y(t) = 2t - 1 + 4e^{-t}$ does satisfy $y' = 2 - 4e^{-t} = -y + 2t + 1$ and $y(0) = 3$.

3. $y'(t) = 2 - 4e^{-t}$. This derivative is negative for $t < \ln 2$, zero at $\ln 2$, positive afterwards. The solution decreases on $[0, \ln 2]$ then increases on $[\ln 2, +\infty[$. Its minimum is $y(\ln 2) = 2 \ln 2 - 1 + 2 = 1 + 2 \ln 2$.

4.
```python
def euler_dm7(N):
    h = 2/N
    y = 3.0
    for k in range(N):
        t = k*h
        y = y+h*(-y+2*t+1)
    return y
```

The time used in the right-hand side is that at the start of the step. The program gives $Y_{10} \simeq 3.42950$ and $Y_{100} \simeq 3.53048$, whereas $y(2) = 3 + 4e^{-2} \simeq 3.54134$. The absolute errors are about 0.11184 and 0.01086. To check these values without a loop, let $Z_k = Y_k - (2kh - 1)$. Euler's recurrence gives $Z_{k+1} = (1 - h)Z_k$ and $Z_0 = 4$. So $Y_N = 3 + 4(1 - \frac{2}{N})^N$.

5. A solution of a differential equation must satisfy an equality between functions on the whole interval, as well as the initial condition. A few numerical values only check a finite number of points; the symbolic substitution of question 2 is a check on the whole domain.

Solutions to the textbook or past-paper work

Exercise 30. If y is constant equal to ℓ, the equation becomes $0 = -\frac{1}{2}\ell + 3$, so $\ell = 6$. Setting $z = y - 6$, we obtain $z' = -\frac{1}{2}z$; all the solutions are $y(x) = 6 + Ce^{-\frac{x}{2}}$, with $C \in \mathbb{R}$.

Exercise 25.

1. $f' = \frac{1}{2}f$ gives $f(x) = Ce^{\frac{x}{2}}$. Since $f(0) = 1$, $C = 1$.

2. $5f' = f$ gives $f(x) = Ce^{\frac{x}{5}}$. The condition $f(1) = 0$ forces $Ce^{\frac{1}{5}} = 0$; since the exponential is never zero, $C = 0$. The solution is the zero function.

3. $f' + f = 0$ gives $f(x) = Ce^{-x}$, so $f'(x) = -Ce^{-x}$. The condition concerns the derivative: $f'(1) = 2$ is equivalent to $-Ce^{-1} = 2$, that is $C = -2e$. Thus $f(x) = -2e^{1-x}$.